GEOMETRIC REGULARITY OF A COMPLETED LOCALISATION AFTER A PURELY INSEPARABLE BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Completion of a finite extension of Noetherian complete local rings is a product of completions. Algebra, Lemma \ref{algebra-lemma-completion-finite-extension}, cited at proof lines 12514-12515. Rewrites the tensor product with L as a product of the completions of B[x] at the primes over q.
- A derivation of a ring extends to a localisation and to a completion. Lemma \ref{lemma-derivation-extends}, cited at proof line 12537. Lets the derivation d/dx act on the completed localisation tensored with K, where it sends x - f to a unit.
- Formal smoothness of the localisation at a maximal ideal and of its completion. Lemmas \ref{lemma-formally-smooth} and \ref{lemma-formally-smooth-completion}, cited at proof lines 12539-12541. Identify the ring tensored with K as formally smooth over A, hence regular.
- A formally smooth algebra over a field is regular, and a quotient of a regular ring by a nonzerodivisor is regular. Proposition \ref{proposition-fs-regular} and Lemma \ref{lemma-quotient-regular}, cited at proof lines 12542 and 12544. Give regularity of the quotient by x - f, which is the ring whose geometric regularity is claimed.

Source reference. Original lemma 07PU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9809–9809. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07N9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9754–9754. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9794–9794. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9442–9442. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9443–9443. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9802–9802. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9795–9795. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let p be a prime number. Let A be a Noetherian complete local domain with fraction field K of characteristic p. Let $\mathfrak{q} \subset A[x]$ be a maximal ideal lying over the maximal ideal of A and let $(0) \neq \mathfrak{r} \subset \mathfrak{q}$ be a prime lying over $(0) \subset A$. Then $A[x]^\wedge_\mathfrak{q} \otimes_{A[x]} \kappa(\mathfrak{r})$ is geometrically regular over $\kappa(\mathfrak{r})$.*

Proof. Note that $K \subset \kappa(\mathfrak{r})$ is finite. Hence, given a finite purely inseparable extension $L/\kappa(\mathfrak{r})$ there exists a finite extension of Noetherian complete local domains $A \subset B$ such that $\kappa(\mathfrak{r}) \otimes_A B$ surjects onto L. Namely, you take $B \subset L$ a finite A-subalgebra whose field of fractions is L. Denote $\mathfrak{r}' \subset B[x]$ the kernel of the map $B[x] = A[x] \otimes_A B \to \kappa(\mathfrak{r}) \otimes_A B \to L$ so that $\kappa(\mathfrak{r}') = L$. Then

$$A[x]^\wedge_\mathfrak{q} \otimes_{A[x]} L = A[x]^\wedge_\mathfrak{q} \otimes_{A[x]} B[x] \otimes_{B[x]} \kappa(\mathfrak{r}') = \prod B[x]^\wedge_{\mathfrak{q}_i} \otimes_{B[x]} \kappa(\mathfrak{r}')$$

where $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ are the primes of $B[x]$ lying over $\mathfrak{q}$, see Algebra, Lemma 07N9. Thus we see that it suffices to prove the rings $B[x]^\wedge_{\mathfrak{q}_i} \otimes_{B[x]} \kappa(\mathfrak{r}')$ are regular. This reduces us to showing that $A[x]^\wedge_\mathfrak{q} \otimes_{A[x]} \kappa(\mathfrak{r})$ is regular in the special case that $K = \kappa(\mathfrak{r})$.

Assume $K = \kappa(\mathfrak{r})$. In this case we see that $\mathfrak{r}K[x]$ is generated by $x - f$ for some $f \in K$ and

$$A[x]^\wedge_\mathfrak{q} \otimes_{A[x]} \kappa(\mathfrak{r}) = (A[x]^\wedge_\mathfrak{q} \otimes_A K)/(x - f)$$

The derivation $D = \mathrm{d}/\mathrm{d}x$ of $A[x]$ extends to $K[x]$ and maps $x - f$ to a unit of $K[x]$. Moreover D extends to $A[x]^\wedge_\mathfrak{q} \otimes_A K$ by Lemma 07PE. As $A \to A[x]^\wedge_\mathfrak{q}$ is formally smooth (see Lemmas 07EC and 07ED) the ring $A[x]^\wedge_\mathfrak{q} \otimes_A K$ is regular by Proposition 07PM (the arguments of the proof of that proposition simplify significantly in this particular case). We conclude by Lemma 07PF. □